بِسۡمِ ٱللَّهِ ٱلرَّحۡمَٰنِ ٱلرَّحِيمِ

Fluid Mechanics

الاسم : احمد جاه الله سادر احمد رقم التسجيل :3900

التخصص: الرياضيات وعلوم الحاسوب

EXERCISES (1)

Question 1

The efficiency η of a fan depends on the density ρ, the dynamic viscosity μ of the fluid, the angular velocity ω, diameter D of the rotor and discharge Q. Express η in terms of the dimensionless parameters (express all powers in terms of powers of Q & viscosity).

Solution: * **Variables ($n = 6$):** $\eta, \rho, \mu, \omega, D, Q$ * **Fundamental Dimensions ($m = 3$):** M, L, T * **Number of π terms:** $k = n - m = 6 - 3 = 3$

Dimensional formulas: * $\eta = M^0L^0T^0$ (Dimensionless) * $\rho = ML^{-3}$ * $\mu = ML^{-1}T^{-1}$ * $\omega = T^{-1}$ * $D = L$ * $Q = L^3T^{-1}$

Repeating Variables: ρ, ω, D

- π_1 **(for η):**

$$\pi_1 = \eta \cdot \rho^a \omega^b D^c$$

Since η is dimensionless, $\pi_1 = \eta$.

- π_2 **(for μ):**

$$\pi_2 = \mu \cdot \rho^a \omega^b D^c$$

$$M^0L^0T^0 = (ML^{-1}T^{-1})(ML^{-3})^a(T^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -1 - b \Rightarrow b = -1$
  - $L: 0 = -1 - 3a + c \Rightarrow 0 = -1 - 3(-1) + c \Rightarrow c = -2$

$$\pi_2 = \frac{\mu}{\rho\omega D^2}$$

- π_3 **(for Q):**

$$\pi_3 = Q \cdot \rho^a \omega^b D^c$$

$$M^0L^0T^0 = (L^3T^{-1})(ML^{-3})^a(T^{-1})^b(L)^c$$

  - $M: 0 = a \Rightarrow a = 0$
  - $T: 0 = -1 - b \Rightarrow b = -1$
  - $L: 0 = 3 - 3a + c \Rightarrow 0 = 3 - 0 + c \Rightarrow c = -3$

$$\pi_3 = \frac{Q}{\omega D^3}$$

Final Answer:

$$\eta = \phi\left(\frac{\mu}{\rho\omega D^2}, \frac{Q}{\omega D^3}\right)$$

Question 2

The resistance force R of a supersonic plane during flight can be considered as a dependent upon the length of the aircraft l, velocity v, air viscosity μ, air density ρ and bulk modulus of air k. Express the functional relationship between these variables and resisting force (express all powers in terms of powers of viscosity and bulk modulus).

Solution: * **Variables ($n = 6$):** R, l, v, μ, ρ, k * **Number of π terms:** $k = 6 - 3 = 3$

Dimensional formulas: * $R = MLT^{-2}$ * $l = L$ * $v = LT^{-1}$ * $\mu = ML^{-1}T^{-1}$ * $\rho = ML^{-3}$ * k (bulk modulus) = $ML^{-1}T^{-2}$

Repeating Variables: ρ, v, l

- π_1 **(for R):**

$$\pi_1 = R \cdot \rho^a v^b l^c$$
$$M^0L^0T^0 = (MLT^{-2})(ML^{-3})^a(LT^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -2 - b \Rightarrow b = -2$
  - $L: 0 = 1 - 3a + b + c \Rightarrow 0 = 1 - 3(-1) + (-2) + c \Rightarrow c = -2$

$$\pi_1 = \frac{R}{\rho v^2 l^2}$$

- π_2 **(for μ):**

$$\pi_2 = \mu \cdot \rho^a v^b l^c$$
$$M^0L^0T^0 = (ML^{-1}T^{-1})(ML^{-3})^a(LT^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -1 - b \Rightarrow b = -1$
  - $L: 0 = -1 - 3a + b + c \Rightarrow 0 = -1 - 3(-1) + (-1) + c \Rightarrow c = -1$

$$\pi_2 = \frac{\mu}{\rho v l}$$

- π_3 **(for k):**

$$\pi_3 = k \cdot \rho^a v^b l^c$$
$$M^0L^0T^0 = (ML^{-1}T^{-2})(ML^{-3})^a(LT^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -2 - b \Rightarrow b = -2$
  - $L: 0 = -1 - 3a + b + c \Rightarrow 0 = -1 - 3(-1) + (-2) + c \Rightarrow c = 0$

$$\pi_3 = \frac{k}{\rho v^2}$$

Final Answer:

$$R = \rho v^2 l^2 \phi\left(\frac{\mu}{\rho v l}, \frac{k}{\rho v^2}\right)$$

Question 3

A partial submerged body is towered in water. The resistance R to its motion depends on the density ρ, the viscosity μ of water, length l of the body, velocity v of the body and acceleration due to gravity g. Show that the resistance to the motion can be expressed in the form:

$$R = \rho l^2 v^2 \phi\left[\left(\frac{\mu}{\rho l v}\right), \left(\frac{lg}{v^2}\right)\right]$$

Solution: * **Variables ($n = 6$):** R, ρ, μ, l, v, g * **Number of π terms:** $k = 6 - 3 = 3$

Dimensional formulas: * $R = MLT^{-2}$ * $\rho = ML^{-3}$ * $\mu = ML^{-1}T^{-1}$ * $l = L$ * $v = LT^{-1}$ * $g = LT^{-2}$

Repeating Variables: ρ, v, l

- π_1 **(for R):**

$$\pi_1 = \frac{R}{\rho v^2 l^2}$$

(Same as previous)

- π_2 **(for μ):**

$$\pi_2 = \frac{\mu}{\rho v l}$$

(Same as previous)

- π_3 **(for g):**

$$\pi_3 = g \cdot \rho^a v^b l^c$$

$$M^0 L^0 T^0 = (LT^{-2})(ML^{-3})^a (LT^{-1})^b (L)^c$$

  - $M: 0 = a \Rightarrow a = 0$
  - $T: 0 = -2 - b \Rightarrow b = -2$
  - $L: 0 = 1 - 3a + b + c \Rightarrow 0 = 1 - 0 + (-2) + c \Rightarrow c = 1$

$$\pi_3 = \frac{lg}{v^2}$$

Final Answer:

$$\frac{R}{\rho l^2 v^2} = \phi\left(\frac{\mu}{\rho l v}, \frac{lg}{v^2}\right) \Rightarrow R = \rho l^2 v^2 \phi\left(\frac{\mu}{\rho l v}, \frac{lg}{v^2}\right)$$

(Proven)

Question 4

The pressure drop ΔP in a pipe of diameter D and length l depends on the density ρ and viscosity μ of fluid flowing at a mean velocity v and the average height of protuberance t. Show that the pressure drop can be expressed in the form:

$$\Delta P = \rho v^2 \phi \left[\frac{l}{D}, \frac{\mu}{vD\rho}, \frac{t}{D}\right]$$

Solution: * **Variables ($n = 7$):** $\Delta P, D, l, \rho, \mu, v, t$ * **Number of π terms:** $k = 7 - 3 = 4$

Dimensional formulas: * $\Delta P = ML^{-1}T^{-2}$ * $D, l, t = L$ * $\rho = ML^{-3}$ * $\mu = ML^{-1}T^{-1}$ * $v = LT^{-1}$

Repeating Variables: ρ, v, D

- π_1 **(for ΔP):**

$$\pi_1 = \Delta P \cdot \rho^a v^b D^c$$

$$M^0L^0T^0 = (ML^{-1}T^{-2})(ML^{-3})^a(LT^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -2 - b \Rightarrow b = -2$
  - $L: 0 = -1 - 3a + b + c \Rightarrow 0 = -1 - 3(-1) + (-2) + c \Rightarrow c = 0$

$$\pi_1 = \frac{\Delta P}{\rho v^2}$$

- π_2 **(for l):** Since l and D have the same dimension, $\pi_2 = \frac{l}{D}$

- π_3 **(for μ):**

$$\pi_3 = \mu \cdot \rho^a v^b D^c$$

$$M^0L^0T^0 = (ML^{-1}T^{-1})(ML^{-3})^a(LT^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -1 - b \Rightarrow b = -1$
  - $L: 0 = -1 - 3a + b + c \Rightarrow 0 = -1 - 3(-1) + (-1) + c \Rightarrow c = -1$

$$\pi_3 = \frac{\mu}{\rho v D}$$

- π_4 **(for t):** Since t and D have the same dimension, $\pi_4 = \frac{t}{D}$

Final Answer:

$$\frac{\Delta P}{\rho v^2} = \phi\left(\frac{l}{D}, \frac{\mu}{\rho v D}, \frac{t}{D}\right) \Rightarrow \Delta P = \rho v^2 \phi\left(\frac{l}{D}, \frac{\mu}{\rho v D}, \frac{t}{D}\right)$$

ASSIGNMENT (I)

Question 1

Using Buckingham's π-theorem show that the velocity through a circular orifice is given by

$$v = \sqrt{2gH}\phi\left[\frac{D}{H}, \frac{\mu}{\rho v H}\right]$$

Solution: * **Variables ($n = 6$):** v, H, D, μ, ρ, g * **Number of π terms:** $k = 6 - 3 = 3$

Dimensional formulas: * $v = LT^{-1}$ * $H, D = L$ * $\mu = ML^{-1}T^{-1}$ * $\rho = ML^{-3}$ * $g = LT^{-2}$

Repeating Variables: ρ, v, H

- π_1 **(for D):** Since D and H have the same dimension, $\pi_1 = \frac{D}{H}$
- π_2 **(for μ):**

$$\pi_2 = \mu \cdot \rho^a v^b H^c$$

$$M^0L^0T^0 = (ML^{-1}T^{-1})(ML^{-3})^a(LT^{-1})^b(L)^c$$

- $M: 0 = 1 + a \Rightarrow a = -1$
- $T: 0 = -1 - b \Rightarrow b = -1$
- $L: 0 = -1 - 3a + b + c \Rightarrow 0 = -1 - 3(-1) + (-1) + c \Rightarrow c = -1$

$$\pi_2 = \frac{\mu}{\rho v H}$$

- π_3 **(for g):**

$$\pi_3 = g \cdot \rho^a v^b H^c$$

$$M^0L^0T^0 = (LT^{-2})(ML^{-3})^a(LT^{-1})^b(L)^c$$

- $M: 0 = a \Rightarrow a = 0$
- $T: 0 = -2 - b \Rightarrow b = -2$
- $L: 0 = 1 - 3a + b + c \Rightarrow 0 = 1 - 0 + (-2) + c \Rightarrow c = 1$

$$\pi_3 = \frac{gH}{v^2} \Rightarrow v = \sqrt{gH} \cdot \text{const}$$

Final Answer: Since $v \propto \sqrt{gH}$, we can write the relation as:

$$v = \sqrt{2gH}\phi\left(\frac{D}{H}, \frac{\mu}{\rho v H}\right)$$

(The $\sqrt{2}$ is a conventional dimensionless constant absorbed into the function).

Question 2

Show that the lift F_L on airfoil can be expressed as $F_L = \rho v^2 d^2 \phi\left(\frac{\rho v d}{\mu}, \alpha\right)$

Solution: * **Variables ($n = 6$):** $F_L, \rho, v, d, \alpha, \mu$ * **Number of π terms:** $k = 6 - 3 = 3$

Dimensional formulas: * $F_L = MLT^{-2}$ * $\rho = ML^{-3}$ * $v = LT^{-1}$ * $d = L$ * $\alpha = M^0L^0T^0$ (Dimensionless angle) * $\mu = ML^{-1}T^{-1}$

Repeating Variables: ρ, v, d

- π_1 **(for F_L):**

$$\pi_1 = \frac{F_L}{\rho v^2 d^2}$$

(Standard aerodynamic force coefficient)

- π_2 **(for α):** Since α is dimensionless, $\pi_2 = \alpha$
- π_3 **(for μ):**

$$\pi_3 = \frac{\mu}{\rho v d}$$

To match the required form, we take the inverse: $\frac{\rho v d}{\mu}$ (Reynolds Number).

Final Answer:

$$F_L = \rho v^2 d^2 \phi\left(\frac{\rho v d}{\mu}, \alpha\right)$$

Question 3

The pressure difference ΔP in a pipe of diameter D and length l due to turbulent flow depends on the viscosity μ density ρ and roughness k using Buckingham's π theorem obtain an expression for ΔP.

Solution: * **Variables ($n = 6$):** $\Delta P, D, l, \mu, \rho, k$ * **Number of π terms:** $k = 6 - 3 = 3$

Dimensional formulas: * $\Delta P = ML^{-1}T^{-2}$ * $D, l, k = L$ * $\mu = ML^{-1}T^{-1}$ * $\rho = ML^{-3}$

Repeating Variables: ρ, μ, D (Choosing μ to focus on viscous effects as typical in pipe flow).

- π_1 **(for ΔP):**

$$\pi_1 = \Delta P \cdot \rho^a \mu^b D^c$$

$$M^0 L^0 T^0 = (ML^{-1}T^{-2})(ML^{-3})^a (ML^{-1}T^{-1})^b (L)^c$$

  - $M: 0 = 1 + a + b$
  - $T: 0 = -2 - b \Rightarrow b = -2$
  - $L: 0 = -1 - 3a - b + c \Rightarrow c = 1 + 3a + b$. Substituting $b = -2$ into M equation: $0 = 1 + a - 2 \Rightarrow a = 1$. Then $c = 1 + 3(1) - 2 = 2$.

$$\pi_1 = \frac{\Delta P \cdot \rho \cdot D^2}{\mu^2}$$

- π_2 **(for l):** $\pi_2 = \frac{l}{D}$
- π_3 **(for k):** $\pi_3 = \frac{k}{D}$

Final Answer:

$$\Delta P = \frac{\mu^2}{\rho D^2} \phi\left(\frac{l}{D}, \frac{k}{D}\right)$$

(Note: This can be rearranged using Reynolds number to match the Darcy-Weisbach equation forms).

Question 4

Derive on the basis of dimensional analysis suitable parameters to represent the thrust developed by a propeller. Assume that the thrust p depends upon the angular velocity ω, speed of advance v, diameter D, dynamic viscosity μ, mass density ρ and elasticity of the fluid medium which can be denoted by speed of sound in medium C.

Solution: * **Variables ($n = 7$):** $p, \omega, v, D, \mu, \rho, C$ * **Number of π terms:** $k = 7 - 3 = 4$

Dimensional formulas: * p (Force) = MLT^{-2} * $\omega = T^{-1}$ * $v = LT^{-1}$ * $D = L$ * $\mu = ML^{-1}T^{-1}$ * $\rho = ML^{-3}$ * $C = LT^{-1}$

Repeating Variables: ρ, ω, D

- π_1 **(for p):**

$$\pi_1 = p \cdot \rho^a \omega^b D^c$$
$$M^0L^0T^0 = (MLT^{-2})(ML^{-3})^a(T^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -2 - b \Rightarrow b = -2$
  - $L: 0 = 1 - 3a + c \Rightarrow c = -4$

$$\pi_1 = \frac{p}{\rho\omega^2 D^4}$$

- π_2 **(for v):**

$$\pi_2 = v \cdot \rho^a \omega^b D^c$$
$$M^0L^0T^0 = (LT^{-1})(ML^{-3})^a(T^{-1})^b(L)^c$$

  - $M: 0 = a \Rightarrow a = 0$
  - $T: 0 = -1 - b \Rightarrow b = -1$
  - $L: 0 = 1 - 3a + c \Rightarrow c = -1$

$$\pi_2 = \frac{v}{\omega D}$$

- π_3 **(for μ):**

$$\pi_3 = \mu \cdot \rho^a \omega^b D^c$$
$$M^0L^0T^0 = (ML^{-1}T^{-1})(ML^{-3})^a(T^{-1})^b(L)^c$$

  - $M: 0 = 1 + a \Rightarrow a = -1$
  - $T: 0 = -1 - b \Rightarrow b = -1$
  - $L: 0 = -1 - 3a + c \Rightarrow c = -2$

$$\pi_3 = \frac{\mu}{\rho\omega D^2}$$

- π_4 **(for C):**

$$\pi_4 = C \cdot \rho^a \omega^b D^c$$

(Same dimensions as v):

$$\pi_4 = \frac{C}{\omega D}$$

Final Answer:

$$p = \rho\omega^2 D^4 \phi\left(\frac{v}{\omega D}, \frac{\mu}{\rho\omega D^2}, \frac{C}{\omega D}\right)$$

Question 5

The resisting force F of a plane during flight can be considered as dependent upon the length of aircraft l, velocity v, air viscosity μ, air density ρ and bulk modules of air k. Expression the functional relationship between these variables and the resisting force using dimensional analysis (NB: unit of k is $ML^{-1}T^{-1}$).

Solution: * **Variables ($n = 6$):** F, l, v, μ, ρ, k * **Number of π terms:** $k = 6 - 3 = 3$ * **Note:** The problem states unit of k is $ML^{-1}T^{-1}$, which is the same as viscosity μ.

Dimensional formulas: * $F = MLT^{-2}$ * $l = L$ * $v = LT^{-1}$ * $\mu = ML^{-1}T^{-1}$ * $\rho = ML^{-3}$ * $k = ML^{-1}T^{-1}$

Repeating Variables: ρ, v, l

- π_1 **(for F):**

$$\pi_1 = \frac{F}{\rho v^2 l^2}$$

- π_2 **(for μ):**

$$\pi_2 = \frac{\mu}{\rho v l}$$

- π_3 **(for k):** Since k and μ have the same dimensions, $\pi_3 = \frac{k}{\rho v l}$

Final Answer:

$$F = \rho v^2 l^2 \phi\left(\frac{\mu}{\rho v l}, \frac{k}{\rho v l}\right)$$